Practical No. 7

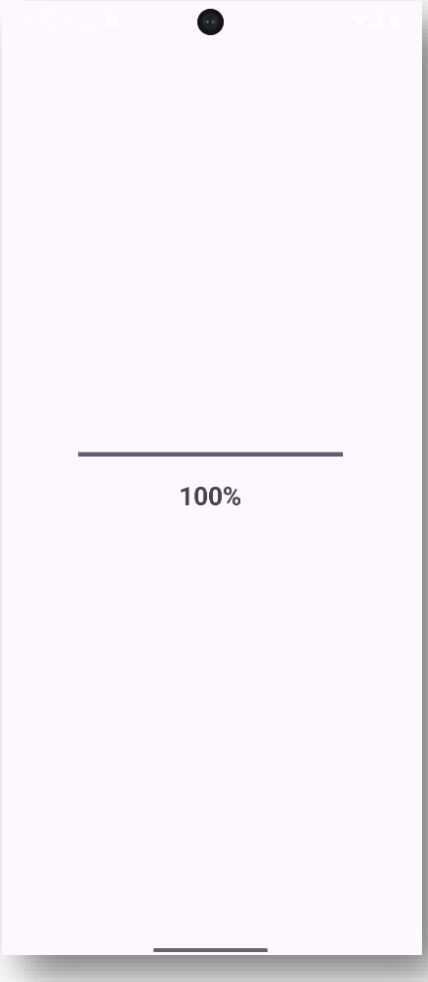
Q.1. Design an Android app that displays a horizontal progress bar updating every second until it reaches 100%.

MainActivity.java :

```
package com.techtitans.madpracticals;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.widget.ProgressBar;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
   int prog = 0;
   @Override

   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      ProgressBar pb = findViewById(R.id.progressBar);
      TextView txt = findViewById(R.id.status);

      Handler h = new Handler(Looper.getMainLooper());

      Thread t = new Thread(() -> {
         while (prog <= 99) {
            h.post(() -> {
               pb.setProgress(prog);
               txt.setText(prog + "%");
            });

          prog++;
            try {
               Thread.sleep(1000);
            } catch (InterruptedException e) {
               e.printStackTrace();
            }
         }
      });
      t.start();
   }
}
```

activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <ProgressBar
        android:id="@+id/progressBar"
        style="?android:attr/progressBarStyleHorizontal"
        android:layout_width="250dp"
        android:layout_height="20dp"
        android:max="100"
        android:progress="0" />

    <TextView
        android:id="@+id/status"
        android:layout_marginTop="12dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="0%"
        android:textSize="24sp"
        android:textStyle="bold" />

</LinearLayout>
```



Output :-

Q.2.Implement a circular indeterminate progress bar that appears during data loading.

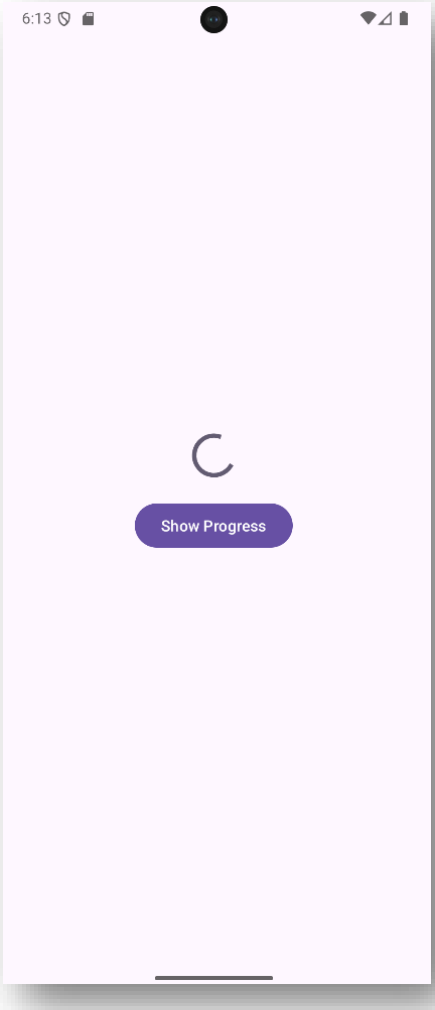
MainActivity.java :

```
package com.techtitans.madpracticals;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ProgressBar;

import androidx.activity.EdgeToEdge;
```

```
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    int prog = 0;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ProgressBar p = findViewById(R.id.progressBar3);
        Button b = findViewById(R.id.button); b.setOnClickListener(v -> {
            p.setVisibility(View.VISIBLE);
        });
    }
}
```

activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <ProgressBar
        android:id="@+id/progressBar3"
        style="?android:attr/progressBarStyle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:visibility="invisible"/>

    <Button
        android:id="@+id/button"
        android:layout_marginTop="16dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Progress" />

</LinearLayout>
```



Q.3.Create a file download simulation using a progress bar and Handler.

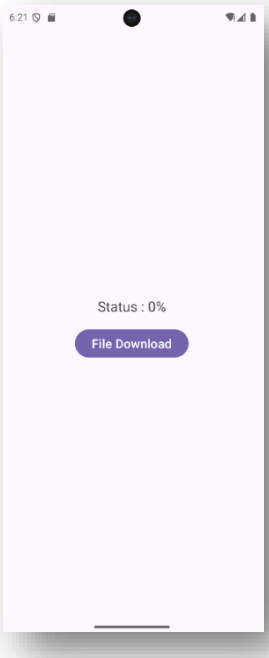
MainActivity.java :

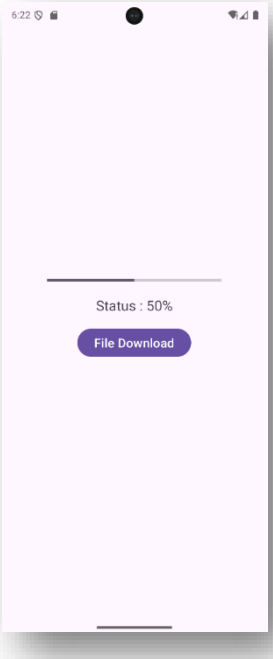
```
package com.techtitans.madpracticals;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.view.View;
import android.widget.Button;
import android.widget.ProgressBar;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    int prog = 0;
    ProgressBar progressBar;
    TextView status;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_hor_progress);

        Button b = findViewById(R.id.button2);
        status = findViewById(R.id.status);
        progressBar = findViewById(R.id.progressBar);

        Handler hd = new Handler(Looper.getMainLooper());

        b.setOnClickListener(v -> {
            progressBar.setVisibility(View.VISIBLE);
            prog = 0;

            Thread th = new Thread(() -> {
                while (prog < progressBar.getMax()) {
                    prog += 10;

                    hd.post(() -> {
                        progressBar.setProgress(prog);
                        status.setText("Status : " + prog + "%");
```

```
                });

                try {
                    Thread.sleep(1000);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
            }
        });
        th.start();
    });
  }
}
```

activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="16dp">

  <ProgressBar
    android:id="@+id/progressBar"
    style="?android:attr/progressBarStyleHorizontal"
    android:layout_width="250dp"
    android:layout_height="20dp"
    android:max="100"
    android:progress="0"
    android:visibility="invisible" />
  <TextView
    android:id="@+id/status"
    android:layout_marginTop="12dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Status : 0%"
    android:textSize="20sp" />
  <Button
    android:id="@+id/button2"
    android:layout_marginTop="16dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="File Download"
    android:textSize="18sp" />

</LinearLayout>
```



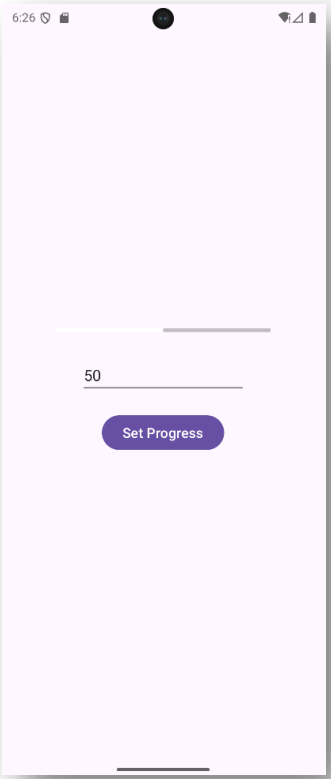


Q.4. Modify the progress bar dynamically based on user input (e.g., slider or button click).
MainActivity.java :

```
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ProgressBar;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Button b = findViewById(R.id.ipval);
        EditText e = findViewById(R.id.progval);
        ProgressBar p = findViewById(R.id.progressBar);

        p.setMax(100);

        b.setOnClickListener(v -> {
            String temp = e.getText().toString().trim();
```

```
            if (temp.isEmpty()) {
                Toast.makeText(this, "Enter value", Toast.LENGTH_SHORT).show();
                return;
            }

            int val = Integer.parseInt(temp);

            if (val < 0) val = 0;
            if (val > 100) val = 100;

            p.setProgress(val);
        });
    }
}
```

activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <ProgressBar
        android:id="@+id/progressBar"
        style="?android:attr/progressBarStyleHorizontal"
        android:layout_width="250dp"
        android:layout_height="20dp"
        android:max="100"
        android:progress="0"
        android:progressTint="@android:color/white"
        android:progressBackgroundTint="#151515" />

    <EditText
        android:id="@+id/progval"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:hint="Enter progress (0–100)"
        android:inputType="number"
        android:textSize="18sp" />

    <Button
        android:id="@+id/ipval"
        android:layout_marginTop="20dp"
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Set Progress"
    android:textSize="16sp" />

</LinearLayout>
```